

HARNESSING EC2, VPC & MOBAXTERM FOR CUTTING-EDGE ENTITY HOSTING

1. PROJECT OVERVIEW

The “Harnessing EC2, VPC & MobaXterm for Cutting-Edge Entity Hosting” project is a cloud-based web hosting solution developed using Amazon Web Services. The main objective of this project is to deploy and host a website securely using AWS EC2 instances within a Virtual Private Cloud environment.

This project demonstrates how cloud infrastructure can be used to create scalable, reliable, and secure hosting environments for organizations and businesses. The deployment process includes configuring EC2 instances, managing secure SSH access through MobaXterm, installing and configuring Apache web server, and hosting a live website publicly using the EC2 public IPv4 address. The project uses Ubuntu Server as the operating system and Apache as the web server software. MobaXterm is used for remote access and secure file transfer to the cloud server. This project provides a practical implementation of cloud computing concepts including server management, VPC security, remote administration, and website deployment using AWS infrastructure.

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2. OBJECTIVES

- To understand and implement AWS cloud hosting services
- To create and configure Amazon EC2 instances for web hosting
- To configure secure VPC networking and security groups
- To establish secure SSH connectivity using MobaXterm
- To install and configure Apache web server on Ubuntu
- To deploy and host a website on AWS EC2
- To manage website files using SFTP functionality
- To understand cloud scalability and secure hosting concepts
- To demonstrate real-world cloud deployment workflow

3. KEY FEATURES

- Cloud-based website hosting using AWS EC2
- Secure SSH remote access using MobaXterm
- Website deployment using Apache Web Server
- VPC-based secure network configuration
- Public website accessibility using EC2 IPv4 address
- File transfer using SFTP
- Scalable and flexible cloud infrastructure
- Linux server management using Ubuntu

4. USE CASE SCENARIOS

Scenario 1: Secure Internal Hosting

Organizations can host internal applications securely within AWS VPC environments while restricting unauthorized access.

Scenario 2: High Availability Web Hosting

Businesses can deploy websites on EC2 instances to handle dynamic traffic efficiently using scalable cloud infrastructure.

Scenario 3: Start-Up Application Deployment

Start-ups can deploy web applications quickly and cost-effectively using EC2 cloud hosting services.

Scenario 4: Educational Cloud Deployment

Students and learners can use AWS EC2 hosting projects to understand practical cloud deployment and Linux server administration.

5. TECHNICAL APPROACH

Frontend (User Interface)

- Developed using HTML and CSS
- Displays project details and cloud deployment workflow
- Provides a clean and responsive website interface

Backend / Server Environment

- Ubuntu Server Operating System
- Apache Web Server for hosting website content
- Linux-based server administration

Cloud Infrastructure

- Amazon EC2 for virtual server hosting
- Amazon VPC for network isolation and security

Remote Management

- MobaXterm for SSH connectivity
- SFTP functionality for file transfer
- Secure PEM key authentication

System Architecture

- Infrastructure Layer → AWS EC2 and VPC
- Server Layer → Ubuntu and Apache Web Server
- Management Layer → MobaXterm and SSH
- Presentation Layer → HTML/CSS frontend

6. IMPLEMENTATION PLAN

Phase	Activities	Timeline
Requirement Analysis	Define hosting objectives and workflow	2 Days
AWS Setup	Configure AWS account and EC2 instance	2 Days
Network Configuration	Configure VPC and security groups	2 Days
MobaXterm Setup	Configure SSH and remote access	1 Day
Web Server Installation	Install Apache on Ubuntu server	1 Day
Website Development	Create website using HTML/CSS	3 Days
Website Deployment	Upload files and host website	2 Days
Testing & Optimization	Test website functionality and security	2 Days

7. BENEFITS

FOR USERS

- Fast and reliable website accessibility
- Secure hosting environment
- Easy remote server management

FOR ORGANIZATIONS

- Scalable cloud infrastructure
- Reduced hardware dependency
- Cost-effective deployment solution

FOR DEVELOPERS

- Hands-on experience with AWS cloud services
- Practical Linux server administration knowledge
- Understanding of real-world deployment workflow

8. PROJECT FLOW

- Create AWS account and configure EC2 instance
- Configure VPC and security groups
- Generate PEM key pair for authentication
- Connect to Ubuntu EC2 instance using MobaXterm
- Install Apache web server on Ubuntu
- Develop website using HTML and CSS
- Upload website files using SFTP
- Configure permissions and restart Apache
- Launch website using EC2 public IPv4 address
- Monitor and optimize server performance

9. REQUIREMENTS SPECIFICATION

System Requirements

- AWS Account
- Ubuntu EC2 Instance
- Minimum 4GB RAM recommended
- Internet Connection
- Web Browser (Chrome, Edge, Firefox)

Software Requirements

- AWS EC2
- Ubuntu Server
- Apache Web Server
- MobaXterm
- HTML and CSS

Installation Steps

Install Apache:

```
sudo apt update
```

```
sudo apt install apache2
```

Start Apache:

```
sudo systemctl start apache2
```

Check Apache Status:

```
sudo systemctl status apache2
```

10. RISKS AND MITIGATIONS

Risk	Impact	Mitigation Strategy
Incorrect Security Group Rules	Website inaccessible	Configure HTTP and SSH properly
Server Downtime	Website unavailable	Monitor EC2 instance regularly
Permission Errors	Files not accessible	Configure proper file permissions
Public IP Changes	Website link changes	Use Elastic IP
Unauthorized Access	Security issues	Use PEM key authentication

11. FUTURE ENHANCEMENTS

- Custom domain integration
- SSL certificate implementation using HTTPS
- Auto Scaling integration
- Load Balancer implementation
- CI/CD deployment pipeline
- Database integration using Amazon RDS
- CloudWatch monitoring dashboard

12. CONCLUSION

The “Harnessing EC2, VPC & MobaXterm for Cutting-Edge Entity Hosting” project successfully demonstrates the practical implementation of cloud computing and website deployment using AWS infrastructure. By integrating Amazon EC2, VPC networking, Ubuntu server administration, Apache web hosting, and secure remote management through MobaXterm, the project provides a scalable, secure, and efficient hosting environment.

This project highlights the importance of cloud technologies in modern web hosting and provides valuable hands-on experience in server deployment, cloud infrastructure management, and Linux-based hosting solutions. With future enhancements such as SSL security, load balancing, and automated scaling, this project can be extended into a fully production-ready cloud hosting platform.